Luke Braithwaite

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Education

University of Cambridge (Peterhouse) Oct 2023 – Present
MPhil Advanced Computer Science, Distinction (Expected) *Cambridge, UK*

- **Modules taken:** Machine Learning and the Physical World, Principles of Machine Learning Systems, Machine Visual Perception, Theory of Deep Learning and Geometric Deep Learning.
- **MPhil thesis:** 'Sheaf (hyper)graph methods for heterogeneous data' supervised by Prof. Pietro Liò.

University of Manchester Sep 2020 – Jun 2023
BSc. (Hons) Computer Science, 1st (80.4%) *Manchester, UK*

- **Dissertation:** 'Bias and Variance in Machine Learning' supervised by Prof. Gavin Brown. **Grade:** 82%.
- **Key modules:** Data Science, Machine Learning, Natural Language Processing, Natural Language Understanding and Mathematical Topics in Machine Learning.
- **First year modules:** Computer Engineering, Computer Architecture, programming in Python and Java, Data Science and Fundamentals of Computation.
- **Second year modules:** Databases, Machine Learning, Programming Languages and Paradigms, Algorithms and Data Structures, and Software Engineering.
- **Third year modules:** Natual Language Understanding, Natural Language Processing and Mathematical Topics in Machine Learning.

Research Projects

Sheaf (hyper)graph methods for heterogeneous data Dec 2023 – Present

- MPhil Advanced Computer Science thesis supervised by Prof. Pietro Liò
- Explored how sheaf-based methods improve performance on heterogenous graph data.

Federated Inferencing for Score-Based Diffusion Models Dec 2023 – Jan 2024

- Developed a novel federated inferencing framework for score-based image diffusion models.
- Implemented approach in PyTorch using the Diffusers library.
- Performed theoretical analysis to gain greater insight of the process

Gaussian Processes for Air Pollution Modelling Dec 2023 – Jan 2024

- Used gaussian processes to estimate air pollution in Greater London.
- Developed the experimental code and analysed the results.

Improving 3D representation with knowledge distillation Nov 2023 – Dec 2023

- Explored how knowledge distillation techniques can improve the performance of 3D scene representations.
- Demonstrated that knowledge distillation is a viable approach for these neural representations.
- Developed a new knoweldge distilled learning approach using the S3IM loss from liteature.

Bias and Variance in Machine Learning Sep 2022 – Jun 2023

- https://aspiecoder1.github.io/uploads/Bias_and_Variance_in_Machine_Learning.pdf
- Third year project/dissertation for my undergraduate degree supervised by Prof. Gavin Brown.
- Explores the effect of bias-variance to explain the behaviour of ML models in a formal manner.
- Performed ML experiments in order to connect the ideas back to published literature and papers.
- Demonstrated that with slight modifications the bias-variance tradeoff remains a useful framework to explain generalisation dynamics of ML models.

Experience

Casual Project Assistant (Software Engineer) Aug 2023 – Oct 2023
The University of Manchester (Imago Software) *Manchester, UK*

- Developed microservices for a Class 2 (Software) medical application that used ML to predict hyperglycemia episodes.
- Deployed microservices to AWS using Elastic Container Services, and automated the deployment to increase development speed and productivity.

Dev Team Member Sep 2022 – Jul 2023
UniCS (The University of Manchester Computer Science Society) *Manchester, UK*

- Designed, developed and manage CS society applications and websites.
- Implemented new CMS systems to improve efficiency of page updates.

- Mananged and refactored in-house hackathon management system.

Graduate Software Engineer (Intern) — Data and Machine Learning Aug 2023 – Oct 2023
THG *Manchester, UK*

- Intern on the machine learning team.
- Developed a statistical simulation of server usage for the cloud services division. From this I estimated stock on hand over time and expected profits based on initial starting conditions.
- Developed a tranformer-based product recommendation system using the NVIDIA Merlin framework and PyTorch.

Casual Project Assistant (Software Engineer) Jan 2022 – Jun 2022
The University of Manchester (Imago Software) *Manchester, UK*

- Laid the foundations for open sourcing a research application for automating data cleaning produced by the Department of Computer Science.
- Documented a large legacy Java codebase from the scratch.
- Designed and planned an application rearchitecture to support a component pipeline-based approach with support for data streams as well as refactoring the code to Kotlin.

Freelance Software Engineer Jun 2021 – Jan 2022
MEF ltd. *Manchester, UK*

- Sole developer on a data-driven application to help economic students visualise economic and climate data and see the impact of economic policies.
- Decreased the initial load time by 40% by refactored the existing application to a be served cache-first using a PWA.

Technical Skills

Languages: Python, PyTorch, JavaScript, TypeScript, ReactJS, Node.js
Software: Git, PyCharm, GitLab, GitHub
Frameworks: PyTorch Geometric

Additional

Rowing (2nd seat) Oct 2023 – Present
Peterhouse Boat Club *Cambridge, UK*

- Noviced for my college during Michaelmas 2023 and continue to row for M3.

Area Umpire (Hockey) 2017 – Present
England Hockey *UK*

- Level 2 outdoor umpire and on the area premier panel for both genders.
- Regularly umpire the East Men's and Women's Premier Divisions.
- Member of the National Young Umpire Promising List, where I umpire nationally and internationally at junior level.